

EnPoWER Insti-Assist — Project Write-Up

1. Chosen Scope and Why

Club Assistant for EnPoWER the Engineering Oriented Promotion of Work Experience and Research initiative, IIT Bombay's undergraduate research-promotion council. This scope was chosen over a general insti-wide assistant for two reasons. First, EnPoWER's own documents (work reports, selection assignments) are specific, and have facts like programme names, dates, coordinator structures that are exactly the kind of thing a general LLM would hallucinate rather than know.

2. Data Sources Used

Seven source documents

- End-Term Work Report, 2023-24 (PDF)
- End-Term Work Report, 2024-25 (DOCX)
- End-Term Work Report, 2025-26 — Final (PDF)
- Mid-Term Work Report, 2025-26 (PDF)
- Academic Coordinator selection assignment, Round 2 / UGAC (PDF)
- Department Research Coordinator (DRC) selection assignment, 2026-27 (DOCX)
- IIT Bombay Students' Gymkhana Constitution, March 2018 (PDF, 81 pages) — the governance document EnPoWER and all other councils operate under

3. Chunking Strategy and Why

Extraction is provenance-first: PDFs are split page-by-page with pdfplumber, and DOCX files are split by Word heading styles falling back to one block if a document has no headings, as with the DRC assignment), plus a separate pass that pulls table rows out as their own units. Every extracted unit carries its exact source file name and page/section label from the start, rather than trying to reconstruct that mapping later.

On top of that, a paragraph-packing chunker merges paragraphs into ~900-character windows (roughly 200-250 tokens) with a 150-character overlap between consecutive chunks from the same page. Three deliberate choices: (1) chunk on paragraph boundaries, never mid-sentence, so a chunk always reads as a coherent unit; (2) never merge text across two different pages into a single chunk, this costs some packing efficiency but keeps every chunk's source citation exact instead of an approximate range; (3) the overlap prevents an answer that happens to sit right on a chunk boundary from being split across two chunks and under-retrieved.

4. Known Limitations / What We'd Improve With More Time

- **No labeled eval set.** The similarity gate was set by manual inspection of a handful of queries, not tuned against a labeled set of in-scope vs. out-of-scope questions. A proper eval e.g. 30-50 questions labeled answerable/unanswerable would let us tune this threshold and measure precision on refusals directly
- **No OCR fallback.** All seven source PDFs are text-native, so this wasn't an issue here, but a scanned or image-only page would silently extract as empty text today.
- **DOCX section granularity is uneven.** The 2024-25 end-term report is well headinged and chunks cleanly; the DRC assignment doc has no heading styles.

- **No multi-turn memory yet.** Each query is currently answered independently; follow-up questions aren't resolved against conversation history.
- **Single embedding model, no hybrid search.** Dense retrieval alone can miss exact keyword/acronym matches (e.g. "SIG", "DRC", "ISAA"); adding a keyword pass alongside the dense one and combining scores would likely help on acronym-heavy queries specifically.